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Digital transformation in the management of higher education institutions

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ABSTRACT

Digital transformation is strategically understood as the process of integrating technological innovations to achieve benefits such as operational efficiency, enhanced customer experience, and business model innovation. Like other sectors, higher education institutions (HEIs) are increasingly interested in this process due to the need to adapt to new business and management models. This transformation entails the strategic exploration of emerging technologies to improve performance, streamline operations, and automate processes, aligning with a society and marketplace that are increasingly technology driven. This approach allows HEIs to remain competitive and relevant. However, it also presents significant challenges for the sector. digital transformation, while inherently complex, often emphasizes specific applications without a holistic understanding of its impact on institutional management. This study seeks to examine the existing body of research on the implementation of digital transformation in HEI management. To achieve this, the study employed the Literature Grounded Theory method. Initially, a bibliometric analysis was conducted to assess the current state of the literature broadly. This was followed by a content analysis to explore the specific characteristics of digital transformation within the context of HEI management. The study identified and categorized 20 distinct technologies used in this process, outlined four categories of challenges associated with their implementation, systematized a set of critical success factors for the transformation of HEIs, and identified key analytical dimensions for transformation within this sector. This structured overview lays the groundwork for new research directions in digital transformation in HEI management.

1. Introduction

Digital transformation is understood as a process that strategically integrates technological innovations, driving significant changes across various sectors and organizations [1]. This process encompasses a broad spectrum of technologies, including Artificial Intelligence, Big Data, Cloud Computing, the Internet of Things, and cyber-physical systems, among others [2]. These technologies can work synergistically, enabling real-time data collection, analysis, and interpretation, thereby fostering automation, enhancing process efficiency, and improving decision-making [3]. The growing interest in digital transformation is fueled by rapid technological advancements, which compel organizations to adapt in order to remain competitive [4]. Additionally, the integration of digital technologies enhances operational efficiency, improves customer experience, drives business model innovation, and enables the achievement of a sustainable competitive advantage [5–7]. The potential benefits, including increased agility, flexibility, and responsiveness to market demands, have attracted significant attention and adoption across various sectors [4,7,8].

Like other sectors, higher education has the opportunity to harness emerging technologies to enhance performance, automate processes, and adapt to a society increasingly driven by technological innovation [9]. This shift has become a priority for Higher Education Institutions (HEIs) [10] as they recognize the crucial role of digital transformation in improving efficiency, fostering industry collaboration, and reducing costs and errors in educational management [11]. Moreover, digital transformation is fueling the rise of "EdTechs"—companies that employ advanced technology in administrative processes and educational systems—thereby intensifying competition with HEIs through greater mobility, flexibility, and extensive use of sophisticated digital platforms

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[12]. These changes are impacting HEIs [13], creating a disruptive environment shaped by new business models [14], where adapting to this phenomenon is essential for institutions to stay relevant [15,16]. In this context, digital transformation is vital for HEIs to maintain a competitive edge [17].

While the importance of digital transformation for HEIs is widely recognized, implementing these strategies requires a cautious approach to developing plans that not only enhance the higher education ecosystem [11] but also maintain competitiveness [15,18]. To achieve this, numerous challenges must be addressed, including failures in project prioritization, decentralized decision-making, internal resistance, and the need to improve employees' digital literacy [10]. Additionally, there may be a lack of alignment in vision, capacity, and commitment necessary for effective implementation [19], as well as the absence of a clear organizational strategy, competing priorities, limited resources, inadequate leadership, and insufficient organizational agility [20]. A poorly managed digital transformation process can have detrimental effects on the profitability, data security, and customer privacy of these institutions [21].

Successfully implementing digital transformation in HEIs requires a strategic approach that considers the unique characteristics and goals of each institution [22] across all areas [23]. This is especially important because (HEIs) are complex organizations that rely on equally complex information systems to manage diverse domains, including academic organization, finance, teaching, and research [24]. Additionally, the execution of plans and strategies involving students, staff, and faculty must be carefully managed [12–16,25].

The development of digital strategies by Higher Education Institutions (HEIs) in response to the growing adoption of new technologies is not a recent phenomenon. Although systematic reviews have extensively examined various aspects of digital transformation in higher education institutions (HEIs), including its challenges and strategic implementations [17,22,26–28], e-learning adoption and technological integration and the development of digital competencies for academic stakeholders [29–32], sustainability-oriented digitalization [33], and digital leadership and digital maturity models [34,35], a critical gap remains in the literature regarding digital transformation in HEI management. Although previous research highlights the technological advancements and pedagogical changes enabled by the adoption of digital technologies, there is still a need for a more comprehensive understanding of digital transformation in the context of HEI management, considering the specific characteristics of this sector.

A more targeted review is essential to analyze how this process unfolds in the administration of these institutions, investigating its implications for governance, decision-making, and operational efficiency. This topic becomes particularly relevant given the increasing complexity of higher education management, the need for data-driven decision-making, and the pressure for greater institutional agility in a scenario of rapid technological changes and economic uncertainty. In this context, conducting a Systematic Literature Review (SLR) emerges as a structured method to consolidate this knowledge, providing an in-depth analysis of the strategies adopted and the challenges faced by HEIs in implementing digital transformation in their management.

Despite the growing body of literature on digital transformation in higher education institutions critical research gaps remain. Most existing studies focus on the adoption of specific technologies or their pedagogical applications, often neglecting the strategic dimensions of digital transformation within HEIs management. This limited perspective overlooks the complexities involved in integrating digital technologies into the unique organizational, cultural, and operational frameworks of HEIs. Moreover, the absence of structured, tailored frameworks hinders these institutions from effectively navigating the transformation process, aligning digital initiatives with institutional objectives, and adapting dynamically to evolving challenges.

To address these gaps, this study investigates digital transformation in the management of higher education institutions, focusing on the key technologies driving this shift, the challenges and barriers faced during implementation, the academic support available for technology adoption, and the critical success factors essential for effective transformation. Thus, this study addresses the following research questions:

- What are the main technologies driving digital transformation in the management of higher education institutions?
- What challenges and barriers do higher education institutions face in implementing digital transformation in management?
- To what extent do existing studies support the implementation of technologies for digital transformation in HEI management?
- What are the critical success factors for digital transformation in HEI management?

While the development of digital strategies by HEIs in response to the growing adoption of new technologies is not a recent phenomenon, there is still a pressing need for a comprehensive understanding of digital transformation within the management of these institutions, particularly in identifying the distinctive characteristics of this sector. Mapping the field through a Systematic Literature Review (SLR) provides a pathway to acquiring this knowledge. Therefore, our article seeks to: (1) Identify and systematize the main technologies driving digital transformation in the management of higher education institutions; (2) Analyze the challenges and barriers faced by HEIs during the implementation of digital transformation in management; (3) Identify the extent to which there are studies that support the implementation of technologies for digital transformation in the management of HEIs; (4) Identify and systematize the critical success factors for digital transformation in HEI management.

This study critically examines research focused on HEI management, identifying the technologies implemented in digital transformation, the critical success factors, and outlining guiding principles for their successful application. To achieve this, both scientometric and bibliometric analyses of the research corpus were conducted. The former evaluates scientific production [36], while the latter examines the evolution of the literature and the dynamics of knowledge dissemination within a specific field over time. It analyzes data extracted from academic databases, considering elements such as citations, authors, keywords, and the diversity of journals reviewed [36]. Within this research context, critical success factors are explored, including the strategies employed by higher education institutions during digital transformation to maintain or enhance their competitiveness. Subsequently, through Content Analysis [37], the study identifies the adoption of 20 technologies within the transformation process and reveals insights that contribute to understanding the primary obstacles faced by these organizations.

The research was carried out using the Literature Grounded Theory method [36], which will be detailed in Section 2 of this article. The Literature Grounded Theory (LGT) is structured within a conceptual model composed of four interconnected stages that guide the review of existing knowledge with the aim of generating new insights. In the first stage, empirical evidence aligned with the research objectives is reviewed, providing the initial foundation for knowledge production. Next, this evidence is systematically decomposed and analyzed to identify relationships among its constituent elements. In the third stage, these relationships are reorganized and synthesized, resulting in the formulation of integrated and innovative knowledge. The method is operationalized through six main stages: (1) defining the scope and review strategy, (2) selecting the corpus for analysis, (3) decomposing and analyzing the elements, (4) synthesizing them into a new perspective, (5) reporting the results, and (6) periodically updating the study to ensure its relevance. Thus, LGT facilitates the identification of research gaps and opportunities while advancing the construction and validation of theories through the structured and consistent integration of empirical and theoretical evidence [36].

Section 3 offers an overview of the digital transformation phenomenon, while Section 4 summarizes the discussion of the results. Finally, Section 5 presents the concluding remarks.

2. Methodological procedures

The SLR provides a comprehensive understanding of the existing research landscape, enabling the identification of research opportunities and emerging trends [38]. Given its configurative nature, where scientific knowledge is analytically generalizable through the synthesis of existing studies, the Literature Grounded Theory (LGT) method was employed. The Literature Grounded Theory (LGT) was chosen for its structured and iterative approach, which extends beyond merely reviewing and analyzing existing literature to facilitate the synthesis and generation of new knowledge. While traditional review methods primarily focus on identifying and organizing available knowledge, LGT

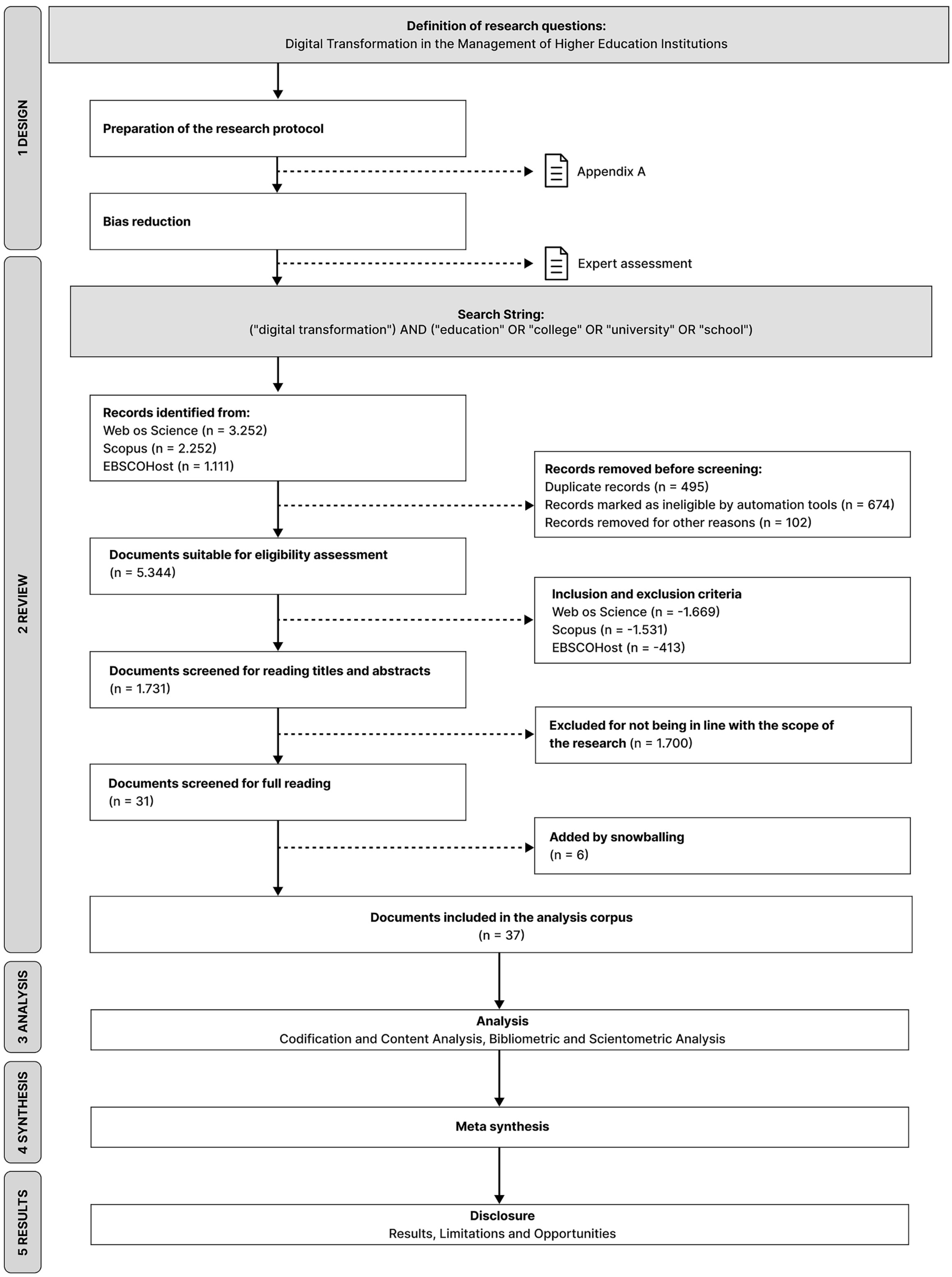

Fig. 1. Logic for conducting a systematic literature review.

takes a more analytical and constructive approach, emphasizing the decomposition, analysis, and synthesis of scientific evidence to foster a deeper and more integrative understanding of the studied phenomenon [36].

This method is structured into five main stages: design, review, analysis, synthesis, and results [36]. These stages involve defining the review's scope and strategy, selecting a corpus for analysis, breaking down and understanding the components, integratively reorganizing these elements, reporting the findings, and continuously updating the research over time [36]. Noteworthy studies that have used this method include research on vaccine development in response to global health threats [39], a study identifying initiatives that support decision-making for adopting energy efficiency measures in production systems [40], and research on the academic software workflow for reviewing, analyzing, and synthesizing literature [41]. Fig. 1 outlines the structure of the present study.

Following the development of the methodological design, a protocol was created to mitigate potential biases. This protocol outlines the data collection strategy, its configuration, and the underlying theoretical framework. Detailed information can be found in Appendix A. The protocol was then validated by experts in the field, chosen for their expertise in digital transformation and/or their involvement in relevant and recent SLR publications. Table 1 lists the experts who evaluated the protocol.

It is important to highlight that specialists 2 and 4 are active professionals in the Information Technology sector, holding strategic and managerial roles in the IT departments of their respective institutions. With the protocol validated, systematic searches were conducted across the defined databases using established criteria and keywords. The articles then underwent an initial screening, during which inclusion and exclusion criteria were applied based on their titles and abstracts. Those selected in this initial phase were thoroughly evaluated, including a complete reading of the texts to assess their relevance, methodological quality, and contribution to the field. Ultimately, a corpus of analysis was compiled, consisting of 37 documents deemed the most relevant and of the highest quality for the review. Details about these selected documents can be found in Appendix B.

In the first stage of the research, Bibliometric Analysis—a set of quantitative procedures used to analyze scientific development over time was employed to describe the field of digital transformation research in terms of growth, structure, interrelations, and productivity. These analyses were conducted using Bibliometrix software in conjunction with the Biblioshiny library, both of which are open-source packages operating within the R statistical environment [42]. The use of these tools aids in identifying prominent authors in the study area, including analyses of authorship, institutional affiliation, and geographical location. Additionally, through its scientometric indicators, the tools enable the evaluation of publication source impact, assessment of productivity, and comprehensive analysis of documents, most-cited words, and identification of researchers most closely aligned with the study, whose work should then be incorporated into the SLR.

To gain a deeper understanding of the current state of research in digital transformation, an additional analysis of the research methods commonly used in the corpus was undertaken to identify methodological trends. This approach helps contextualize the findings and accounts for the limitations and potential biases associated with various methodological approaches. In this study, a thematic map was also utilized, offering indicators for analyzing the evolution of the studied themes. The results were generated by a semi-automatic algorithm that analyzed the abstracts within the corpus [42]. By evaluating the distribution of topics across the quadrants, researchers can identify key research trends in the context of digital transformation in HEI management. Additionally, this approach helps guide the identification of research directions that may gain prominence in the future.

The thematic map is divided into four regions (or quadrants). The first quadrant (Motor Themes) is characterized by high density and contextual centrality, encompassing the core themes that are highly relevant and may indicate promising areas of study. The second quadrant (Niche Themes) includes specific and atypical themes that, despite being highly developed, have low centrality. The third quadrant (Emerging or Declining Themes) signals themes that are losing attention and relevance. Lastly, the fourth quadrant (Basic Themes) covers the most fundamental themes, distinguished by high centrality but low density [43].

Contextual centrality measures the degree of connection a theme has with others within the research field. High centrality indicates a theme that is well-integrated into the knowledge structure, linking different areas. Low centrality suggests a more peripheral theme, which may be either emerging or in decline. Density reflects the internal cohesion and conceptual development of a theme. High density indicates a well-structured and consolidated theme, while low density may suggest a fragmented or developing research area. These two metrics are calculated based on keyword co-occurrence networks extracted from the analyzed content. The centrality score is derived from the sum of connections a given theme has with others in different clusters, while the density score is computed by evaluating the number of connections between keywords within the same cluster [43]. Fig. 2 illustrates the structure of the thematic map, highlighting its quadrants.

The next phase of this study involves coding the analytical corpus. A mixed coding scheme was employed, using frequency as the counting principle, and the coding was conducted in three stages: open coding, axial coding, and selective coding. Open coding entails extracting meaning from the text and encapsulating it into codes. Axial coding involves grouping these codes into categories based on their similarities and/or differences. Finally, selective coding focuses on identifying and establishing relationships between these categories [36]. The coding strategy incorporated both predefined (a priori) and emergent (a posteriori) codes and categories [36]. The predefined codes were selected from recent studies that used similar methodologies and addressed topics related to the research focus of this study. In configurative reviews, which aim to generate or explore a theory, some fundamental concepts (codes) may be predefined (a priori). However, many others emerge during the research analysis (a posteriori) as new interpretations and connections take shape. This approach enhances the understanding of the investigated phenomenon, allowing the theory to evolve iteratively through the critical examination of studies [44]. A priori concepts provide a structured foundation for coding, anchoring the analysis in established theoretical frameworks. However, as the review progresses, new codes may emerge a posteriori, incorporating unforeseen insights and enriching the overall understanding of the subject under

Table 1
List of experts.

Researcher/Expert	Background	Affiliation	Position	Experience (in years)	Country
Expert 1	PhD in Production and Systems Engineering	UNIFAP	Professor and Researcher	10	Brazil
Expert 2	PhD in Information Science	UNIFAP	Professor, Researcher and Research Coordinator	12	Brazil
Expert 3	PhD in Management Science	UCP	Professor and Researcher	9	Portugal
Expert 4	PhD in Computer Science	UFPE	Researcher and Vice-Director	20	Brazil
Expert 5	PhD in Design	CESAR	Research Manager	8	Brazil

Source: Prepared by the authors.

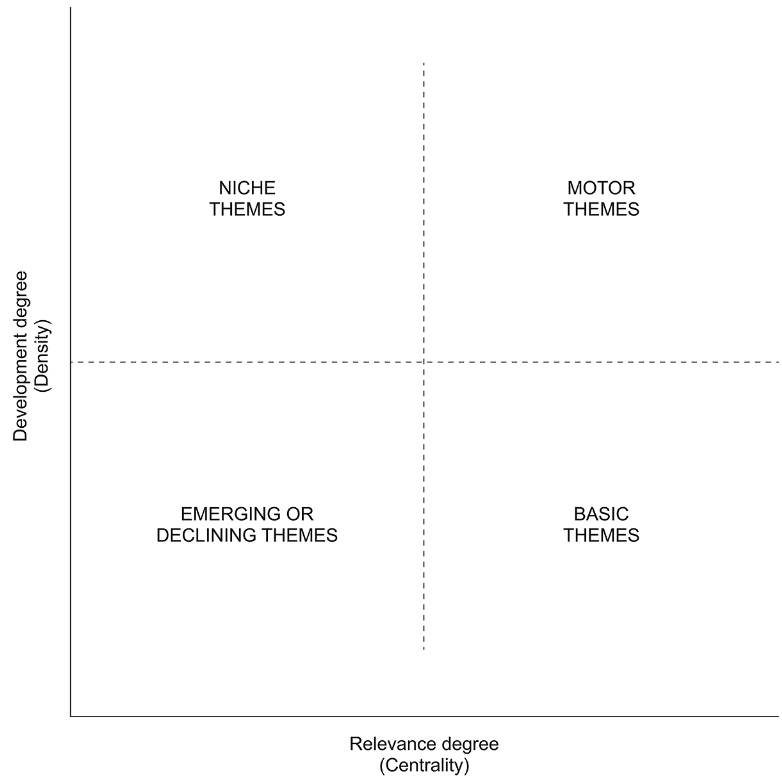

Fig. 2. Thematic map classification structure.

investigation [45]. The selection of codes was guided by relevance, allowing for the inclusion of previously validated analytical categories. The coding process was carried out using Atlas.ti software, and the results, including the identified predefined codes, their descriptions, and types, are systematically presented in Table 2.

This approach allows for a contextualized understanding of the intrinsic elements within the corpus, facilitating the extraction of significant insights and, consequently, the development of a comprehensive understanding of the phenomena explored in the research [37]. After completing the content analysis using the 17 resulting codes, considering 10 a priori codes (Table 2) and 7 a posteriori codes (Table 6), the literature was synthesized to create new knowledge by integrating the disaggregated parts of the corpus [48].

To identify and systematize the technologies adopted by HEIs, two distinct classification categories were employed. First, the technologies were classified according to their functions related to data, as proposed by [2]. Next, these technologies were organized into three groups following the approach of [49]. Table 3 outlines the categories within each classification model.

Meta-synthesis was used to synthesize the results, chosen for its ability to elucidate and explore underlying aspects within a diverse set of

Table 2
Codes created for content analysis in a predefined manner.

Code	Description	Type	Source
Background	Events or conditions that occur before a phenomenon.	A priori	[36]
Barriers or Difficulties in Management	Related difficulties.	A priori	[46]
Strategy	A plan of action designed to achieve specific objectives.	A priori	[46]
Phenomenon	Identifies the phenomenon of interest.	A priori	[36]
Tool, Technique, Method and/or Framework	Instruments or approaches used.	A priori	[47]
Research Method	Approach or strategy adopted to conduct the research.	A priori	[47]
Study Limitation	Restrictions or conditions that affect the generalization of the results.	A priori	[47]
Theoretical Assumption	Fundamental assumptions that guide the research.	A priori	[46]
Research Question	Question formulated to guide the investigation.	A priori	[47]
Result	What is the consequence, the effect of an action.	A priori	[47]

Source: Prepared by the authors.

Table 3
Classification Models for digital transformation Technologies.

Classification Model	Categories	Description
[2]	**Data generation and collection**	Technologies that capture data from the environment and from people.
	Data transmission for storage or use	Technologies that transfer data from its point of origin to the storage or use locations.
	Data conditioning, storage and processing	Technologies that cover the storage, maintenance, availability and transformation of data, generating knowledge.
	Data application	Technologies that apply data, creating value
[49]	**Enabling Technologies**	Fundamental tools and resources that enable the technological infrastructure, promoting the operation of more advanced systems.
	Integration Technologies	Solutions that facilitate the union of different components and systems, promoting interoperability and communication between them.
	Application Technologies	Tools for specific use, applied directly in practical activities, contributing to concrete objectives.

Source: Prepared by the authors based on [2,49].

primary studies [50]. The analysis and synthesis of the literature informed the development of a conceptual model for digital transformation in HEI management, clarifying results, limitations, and research opportunities. Finally, the model proposed by [51] was adopted to organize the analytical corpus based on the thematic focus established by the researchers. The following section discusses the results obtained.

3. Digital transformation

The concept of digital transformation is marked by a lack of consensus regarding both the term itself [52] and the scope it covers [1]. Despite this, there is widespread agreement on the urgent need for companies to adapt to this new paradigm [4,5,7].

In the organizational context, it is important to distinguish between “digitalization” and “digital transformation,” as they are often mistakenly used interchangeably. Digitalization is considered a preliminary stage of digital transformation and refers to the conversion of analog information into digital formats, without necessitating significant revisions to existing processes, business models [53], or organizational practices [54].

Digital transformation, by contrast, goes beyond digitalization by redefining organizational processes, business strategies, and operational models through the strategic and innovative incorporation of digital technologies [55]. In this sense, digital transformation represents not just the adoption of digital technologies, but a reconfiguration of the organization to exploit new opportunities and improve efficiency [10].

Conceptually, digital transformation is defined as “a process of fundamental change enabled by the innovative use of digital technologies, accompanied by the strategic leveraging of essential resources and capabilities, to radically improve an entity and redefine its value proposition for its stakeholders” [51]. In this context, an entity can be an organization, a business network, an industry, or society at large, and digital transformation therefore manifests across six fundamental domains: nature, entity, means, expected outcome, impact, and scope [51]. This methodological approach offers a critical analytical framework for in-depth investigations and a more comprehensive understanding of the evolution of the digital transformation concept. By establishing a common language, the definition proposed by [51] significantly contributes to the consolidation of the field, facilitating deeper discussions, more robust research, and more effective implementations.

4. Analyses OF results

4.1. Scientometric analysis

The scientometric analysis identified the key researchers in the field of digital transformation in HEIs, the journals where their research was published, and the publication timeline of the articles in the corpus. The results of these analyses are presented in Table 4.

A noticeable evolution in the topic is evident, with a rise in publications starting in 2017 and accelerating from 2020 onward. This trend highlights the increasing interest in digital transformation within HEI management. This surge can be attributed to the pressure for digital transformation during the COVID-19 pandemic, which intensified competition in the digital landscape [25], despite research showing that the adoption of digital technologies and processes is inevitable for higher education management [15]. To remain relevant, higher education institutions must adopt business practices and management methods [56,57].

Regarding the methodological approaches (Table 5), the most common research methods in digital transformation within HEI management are Quantitative Analysis (38 %), Literature Analysis and Synthesis (27 %), and Case Studies (19 %). The predominance of Quantitative Analysis reflects a preference for an objective and measurable approach, aimed at validating or generalizing findings across a representative sample. The use of Literature Analysis and Synthesis indicates the importance of establishing theoretical foundations to support discussions on digital transformation. Finally, the adoption of Case Studies can be attributed to the complex nature of digital transformation, which involves organizational, cultural, and technological aspects, making it valuable to examine specific cases, identify patterns, and explore challenges, solutions, and outcomes in real-world situations. However, some studies employ mixed-method approaches, making it difficult to classify them under a single category. In such cases, we have opted to categorize the method explicitly stated by the authors as the primary research method used in the study.

Table 4
Main authors and research sources.

Analysis Dimension	Name	Country	# of publications	Publication period
Researchers	Akhmetshin Em.	Russia	4	2019 – 2020
	Safiullin Mr.	Russia	4	2019 – 2020
	Hashim Mam.	United Arab Emirates	2	2022
	Matthews R.	England	2	2022
	Tlemsani I.	England	2	2022
	Vasilev Vl.	Russia	2	2019 – 2020
	Aditya Br.	Indonesia	1	2021
	Alenezi M.	Saudi Arabia	1	2021
	Alhubaishy A.	Saudi Arabia	1	2021
Source	Education And Information Technologies		5	2017 – 2023
	Sustainability		4	2019 – 2023
	International Journal of Engineering and Advanced Technology		3	2018 – 2023
	Education Sciences		2	2020 – 2023
	International Journal of Information and Education Technology		2	2020 – 2023
	IEEE Frontiers in Education Conference (FIE)		1	2018
	Cataloging & Classification Quarterly		1	2023
	Exploring Services Science		1	2021
	Future Internet		1	2020
	IEEE Access		1	2023

Source: Prepared by the authors.

Table 5
Methodological approaches.

Research Method	Amount	Percentage (%)
Quantitative Analysis	14	37,84
Literature Analysis and Synthesis	12	32,43
Case Study	7	18,92
Grounded Theory	1	2,7
Multilevel Approach	1	2,7
Design Science Research	1	2,7
Computer Modeling and Simulation	1	2,7

Source: Prepared by the authors.

The map (Fig. 3) categorizes the keywords present in the corpus into four regions, highlighting different research focuses on digital transformation in higher education institutions. In the upper right quadrant, labeled Motor Themes, topics such as ’results,’ ’institutions,’ and ’university’ are identified as promising areas, indicating a focus on analyzing outcomes in these institutions. In contrast, the upper left quadrant, Niche Themes, includes frequent topics like ’strategies,’ ’challenges,’ and ’analysis,’ which, despite being frequently discussed, have not shown a significant impact on research. The lower left quadrant, titled Emerging or Declining Themes, points to a lesser exploration of topics related to ’leaders,’ ’support,’ and ’competitive,’ suggesting

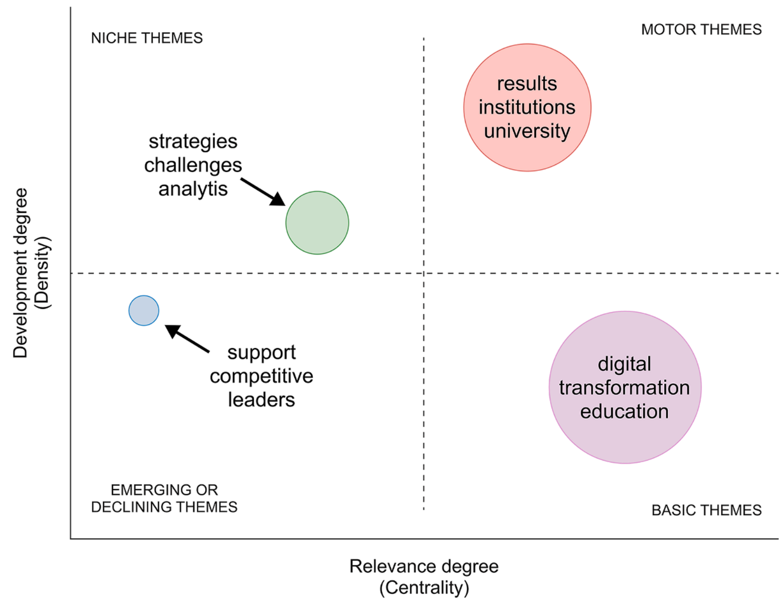

Fig. 3. Thematic map.

that although these topics are still present, they have low centrality in current investigations. Therefore, the theme of strategies to sustain or increase competitiveness may be emerging, indicating the need for more studies to increase the centrality of this topic.

Finally, the lower right quadrant, Basic Themes, encompasses topics associated with digital transformation in education, identified by keywords such as ’digital,’ ’transformation,’ and ’education.’ Although less frequent, these themes indicate potential growth in relevance in future research on the teaching and learning context.

Although topics such as strategies and challenges are frequently discussed, they do not appear to have driven significant advances in research, which may suggest a saturation of investigations in these areas without impactful or innovative contributions. This observation reveals a possible redundancy in academic research, where the frequency of discussion on a topic does not necessarily translate into substantial progress or relevance. Additionally, it was noted that critical areas such as leadership and competitiveness are receiving less attention in current research. This may reflect a misalignment between the emerging needs of higher education institutions and the current focus areas of academic research, potentially indicating a gap in preparing these institutions to face contemporary challenges in the digital landscape.

4.2. *Content analysis*

In the next stage, the study delves deeper into the corpus of analysis. During this phase, new codes emerged that were not initially considered. These new codes broadened the scope of the content analysis, allowing for a more detailed understanding of the object of investigation. The emergent codes are listed in Table 6.

Table 6
Codes created for emergent content analysis.

Code	Description	Type
Definition of DT	Understanding or interpreting the concept of digital transformation.	A posteriori
Dimension	Facets or components that, when integrated, provide a holistic view of the process.	A posteriori
Project Stage	Phases or steps in the development of a digital transformation project.	A posteriori
Critical Success Factor	Fundamental elements for success in digital transformation.	A posteriori
Skill	Skills required to implement digital transformation.	A posteriori
Business Model	Implementation of changes in the business model.	A posteriori

Source: Prepared by the authors.

Understanding the phenomenon within the context of HEIs is essential for identifying the unique characteristics of these organizations. In the literature, digital transformation is described as a continuous technological evolution that drives strategic changes across various organizational aspects, such as infrastructure and operations. Within HEI management, this understanding extends to encompass the specific nuances of the educational environment. digital transformation in HEIs paves the way for a new university model known as the "Digital University," characterized by the advanced integration of digital technologies into its activities, transforming the academic environment to meet the demands of the digital age [58,59]. In terms of management, this involves redefining educational services to accommodate changes in business processes, offering digital education, research support, and services. It also requires reconfiguring the integration of technologies and processes (systems, data, and equipment) with social factors (people and culture), aiming to create enhanced environments for students, faculty, and researchers.

Thus, the concept of digital transformation in HEI management impacts a wide range of dimensions, including university structure, services for students and staff, and technological expansion across all departments [15]. This transformation necessitates continuous investments in professional development, particularly in the field of information technology [60].

A recurring theme in the discussion is the recognition of digital transformation as a comprehensive, strategic, and ongoing process [61]. It involves not only the adoption of new technologies but also demands profound organizational transformations, integrating information, processes, and human elements [22]. This transformation is seen as an effort that transcends technological domains, influencing cultural, organizational, and social dimensions [6]. It is viewed as a strategic shift that integrates digital technologies into every aspect of an organization, impacting its infrastructure, operations, products, and services [62]. The goal is to align the institution with the demands of the digital age, balancing the integration of new practices while preserving the essential elements of existing organizational structures [10]. In essence, digital transformation represents a paradigm shift, particularly in management, introducing new actors, structures, practices, values, and beliefs into the educational ecosystem [63]. As higher education institutions evolve into digital universities, this transformation becomes a strategic imperative, shaping educational services and business processes to meet the dynamic demands of an ever-changing landscape [22].

4.3. Mapping and analysis of technologies present in the corpus

The literature on digital transformation in HEI management identifies twenty different technologies. These technologies encompass a diverse array of tools and platforms that are crucial for modernizing and enhancing both educational and administrative processes. The technologies were organized based on the classifications by [49] and [2], as illustrated in Fig. 4.

IoT, Cloud Computing, Big Data, and Artificial Intelligence are the technologies most frequently cited in the analyzed literature, suggesting a trend toward adopting these as foundational stages in the digital transformation process. The integration of cloud computing offers numerous advantages, including reduced technological infrastructure costs, rapid deployment of virtual computing resources, process improvements through platform and service integration, enhanced mobility, and mechanisms to ensure business continuity. Consequently, building internal competencies in cloud computing through an evolutionary learning process provides a significant advantage in developing a successful digital transformation strategy [3]. Cloud computing has become a driving and transformative force in the business models of universities [64].

The second most prevalent technology is IoT. As its adoption expands from classrooms to student residences, universities must establish robust technical infrastructures to support the interconnection of these devices. By implementing IoT, universities can efficiently monitor and manage resources such as lighting, climate control, and security, resulting in cost savings and enhanced environmental Sustainability [65]. Furthermore, the real-time data collection enabled by IoT allows for more in-depth analysis of academic performance, facilitating the personalization of teaching strategies to meet individual student needs [66]. Thus, the implementation of IoT not only modernizes infrastructure but also serves as a student retention strategy. Providing a connected and technologically advanced academic environment attracts and retains students by offering a comprehensive learning experience [3].

Artificial Intelligence plays a pivotal role in both academic activities (such as learning and research) and management, making it a central component of the digital transformation process within institutions [67]. Its implementation facilitates the deployment of various tools, including chatbots, virtual assistants, dropout risk management systems, student performance tracking, predictive resource analysis, adaptive learning platforms, administrative decision-making, and student admissions processes [68]. AI enables the processing of vast amounts of data, providing advanced analytics to enhance strategic decision-making. By adopting AI-based systems, higher education institutions can personalize services, effectively meeting the specific needs of students, staff, and suppliers. Furthermore, AI supports the creation of customized educational experiences, dynamically adapting to individual student characteristics through machine learning algorithms that analyze performance patterns, thereby informing the development of pedagogical strategies [22] and streamlining processes to enhance flexibility and support the adaptation of business models [69].

Similarly, Big Data solutions have been employed in management to collect and process data, tailoring products and services to the preferences of students and staff. These solutions aid in the reconfiguration of study programs, enhance educational experiences for teachers, students, and researchers, optimize infrastructure, and facilitate the efficient allocation of resources and investments [22]. However, in higher education institutions, the rapid growth in the volume and sources of data poses challenges in collecting and managing these assets to transform them into actionable knowledge for managers. Despite these challenges, these processes are essential to the success of any digital transformation strategy [20].

Thus, the effective integration of these technologies into HEI management can begin by adopting a systemic architecture that unifies the data flows from IoT sensors, Big Data platforms, and Artificial Intelligence applications into a single, cloud-based ecosystem. This setup enables continuous performance analyses through predictive models that anticipate academic and administrative needs, thereby supporting real-time strategic decision-making. By connecting routine automation (e.g., chatbots and virtual assistants) with machine learning mechanisms, institutions can streamline processes, making it possible to personalize services, upgrade infrastructure, and more precisely allocate resources. In addition, the convergence of analytics platforms and large-scale datasets facilitates the identification of dropout patterns, the development of targeted retention strategies, and an integrated approach to managing the student lifecycle.

The technologies mentioned are largely interconnected and possess

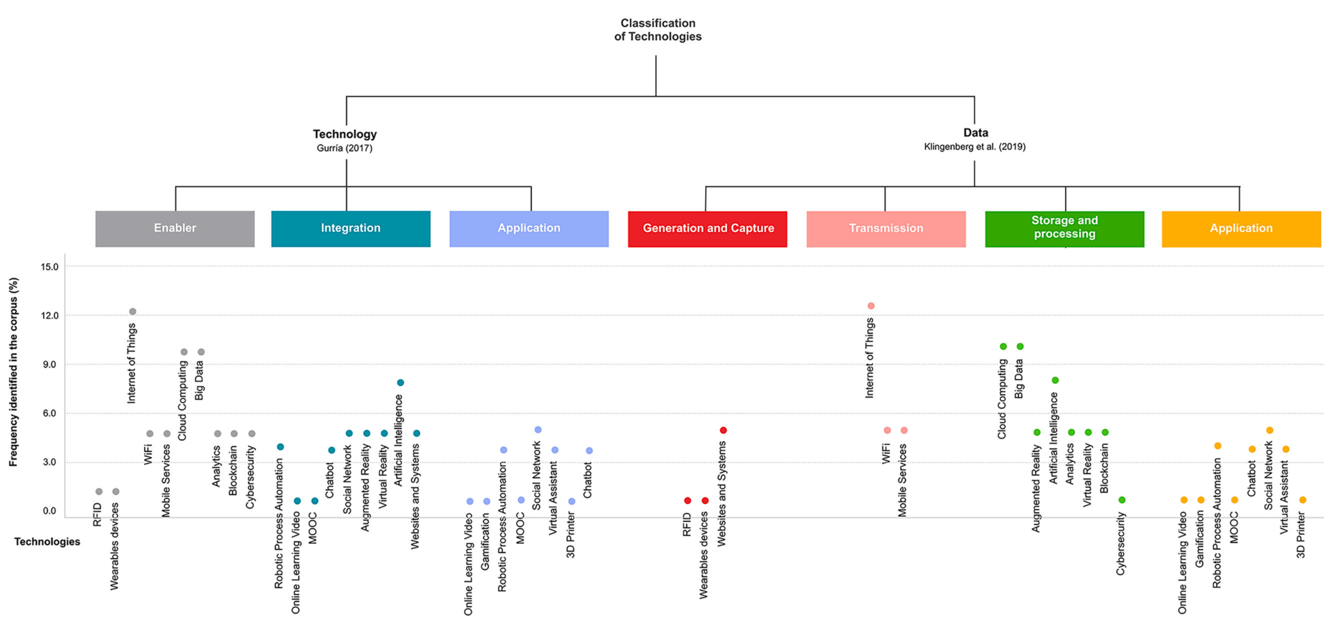

Fig. 4. Classification of digital transformation technologies in HEIs.

an integrative, progressive nature. Therefore, it is essential to analyze how these technologies are used together. To facilitate this, a framework was developed (Table 7), with the horizontal axis displaying the technologies classified according to [49] and the vertical axis showing the classification by [2].

The co-occurrence analysis highlights a focus on Artificial Intelligence (21 occurrences), Cloud Computing (20 occurrences), Analytics (18 occurrences), and Big Data (13 occurrences). These technologies are not only utilized independently but also in combinations that amplify their effectiveness. This underscores that digital transformation in HEIs involves technologies across all classification categories, reflecting a diverse and integrated technological ecosystem.

Analyzing the presence of these technologies in higher education institutions is crucial for several reasons. First, it provides a clear understanding of the key technologies driving digital transformation in this context, equipping managers and decision-makers with the knowledge needed to explore available options. Additionally, it offers guidance for implementing strategies more effectively, tailored to the specific needs and challenges of each institution. Notably, higher education institutions have been utilizing technologies across all classification levels: enablers, integrators, and application technologies. This widespread adoption can be attributed to the extensive changes brought about by digital transformation in this sector, a topic that will be discussed in greater detail.

Despite their promising benefits, these technologies face significant challenges in implementation. The adopted coding methodology identified the main difficulties associated with integrating these technologies into HEI management, thematically grouping them into four categories. This classification was specifically based on the “Barriers or Difficulties in Management” codes. Articles assigned this code were thoroughly analyzed to identify specific barriers, which were then categorized to enhance clarity and provide a structured overview of the challenges faced. Research also highlights considerable challenges related to infrastructure and long-term efficiency. As new functionalities are

Table 7
Technologies implemented in conjunction with other technologies in HEI environments.

			TECHNOLOGY GURRÍA (2017)																				
			ENABLING				INTEGRATING					APPLICATION											
			ANL	AI	BD	CS	BC	CC	IOT	RFID	WIFI	3DP	AR	CB	GF	MS	MOOC	OLV	RPA	SNET	VA	VR	TOTAL
DATA KLINGENBERG ET AL. (2022)	GENERATION AND CAPTURE	RFID **(RFID)**	1						2		1	1											**5**
		Internet of Things **(IOT)**		1		2		3		2		2											**10**
		Mobile Services **(MS)**	2		1						1						1		1	2			**8**
		Virtual Reality **(VR)**											3										**3**
		Augmented Reality **(AR)**																				3	**3**
	TRANSMISSION	WIFI **(WIFI)**	2		1	1		3		1						1	2	2					**13**
		Blockchain **(BC)**				1		3															**4**
	CONDITIONING, STORAGE AND PROCESSING	Cloud Computing **(CC)**	1	2	2	1	3		3		3						1	2	2				**20**
		Big Data **(BD)**	3	2				2			1			1		1		1		2			**13**
		Cybersecurity **(CS)**		1				3	2		1												**7**
	APPLICATION	Analytics **(ANL)**		3	3			1		1	2			1		2	2	2	1				**18**
		Artificial Intelligence **(AI)**	3		2	1		2	1					3	1		1	3	2		2		**21**
		Chatbots **(CB)**	1	3	1																		**5**
		MOOC **(MOOC)**	2	1				1			2					1		1					**8**
		Online Learning Video **(OLV)**	2	3	1			2			2						1						**11**
		Gamification **(GF)**		1																	1		**2**
		RPA **(RPA)**	1	2				2								1					1		**7**
		Social Networks **(SNET)**			2											2							**4**
		Virtual Assistants **(VA)**		2											1				1				**4**
		3D Printer **(3DP)**							2	1													**3**

CAPTION: 0 1 2 3

Source: prepared by the authors.

introduced, digital transformation imposes additional demands on IT infrastructure, leading to an increasingly specialized workload over time. These factors can prolong the institution's adaptation process across all areas, compromising the quality of digital resources, impacting user services and management support tools. Such challenges negatively affect institutional efficiency, user experience, and contribute to increased resistance to change. Table 8 summarizes the main difficulties identified in the primary studies.

Another key aspect highlighted by the analysis involves the dimensions impacted by digital transformation within HEIs. Recognizing these diverse perspectives enables the development of strategies that comprehensively address the various stakeholders involved in this process. This focus underscores the complexity of digital transformation, as reflected in the consensus among authors on the need for a deeper understanding of this phenomenon.

[72] identify three dimensions of analysis: technological, organizational, and social. In contrast, [21] propose four dimensions: economic, social, environmental, and cultural. Similarly, [73] also identify four dimensions: cultural, social, economic, and environmental.

Economically, digital transformation offers benefits such as more profitable business models, streamlined processes, and lower information exchange costs [74]. Socially, it enhances the student experience and lifecycle by improving traditional educational models and reconfiguring the format of education. Environmentally, it emphasizes the adoption of clean technologies, thereby reducing environmental impacts [75]. Culturally, it highlights the importance of connected and adaptive knowledge within the context of digital transformation in higher education, emphasizing the role of networks and connections in the academic environment [69].

Fig. 5 illustrates the convergence of these dimensions in the analysis of digital transformation, integrating the perspectives of [72,21], and [73].

The intersections in Fig. 5 highlight the interdependencies between these dimensions, emphasizing that digital transformation in HEIs is not an isolated process, but rather the outcome of dynamic interactions [6]. For instance, the overlap between technology and culture underscores the need for institutional adaptation and innovation alongside technological adoption [62]. Similarly, the intersection of the social dimension with other areas illustrates how digital transformation shapes student experiences, teaching models, and academic relationships [69]. Though represented separately, the economic and environmental dimensions are closely linked, as digital technologies drive both financial optimization and sustainable practices, directly influencing institutional structure, governance, and operational processes [10].

The operational dynamics of these higher education institutions have been evolving under the influence of technological changes to meet the demands of industry and society [76]. This pressure is driving HEIs to digitize their educational, scientific, innovative, and entrepreneurial activities [76]. As HEIs navigate this digital shift, their transformation strategies can vary significantly depending on institutional goals, available resources, and external pressures. This divergence in approach underscores two primary expected outcomes of the transformation process: economic performance and capability enhancement [51]. In practical terms, some organizations opt for a gradual approach, beginning with specific digitization projects. These projects may involve introducing digital technologies in targeted areas of the organization, laying the groundwork for broader initiatives. Other organizations pursue a more comprehensive approach, striving to achieve digital transformation directly. This approach involves a radical transformation across the organization, encompassing not only the implementation of digital technologies but also the reevaluation and reinvention of processes, organizational culture, and business models to better align with the demands of the digital age [51].

A key point to consider in this analysis is the process of implementing digital transformation within the context of HEI management. The literature underscores the importance of recognizing the unique characteristics of each sector and type of business—a complexity compounded by the absence of a universal model identified in the studies [77]. Another critical insight from this study is that digital transformation necessitates a shift in the business model [78], a complex change that can nonetheless optimize products, services, and operations, enhancing efficiency compared to traditional management approaches [79].

In the context of HEIs, three fundamental factors underpin the development of competitive advantage: a) The industry's attractiveness for achieving long-term profitability, b) The drivers that influence both attractiveness and profitability, and c) The relative positioning of universities within the sector to obtain the authority to confer degrees or professional qualifications [67]. Therefore, effectively leveraging digital transformation in HEIs requires engaging not only all infrastructure components but also the intellectual and cultural resources available [58]. The research reviewed provides guidelines for developing agile, realistic, and scalable digital transformation strategies that serve as the core integrating philosophy across all university functions [67]. This is because the integration of digital technologies enables universities to transcend traditional virtual boundaries, impacting course offerings, shaping delivery methods, and influencing the entire value chain within the university [80,81].

Moreover, leadership is responsible for establishing a shared understanding of the vision and objectives of digital transformation, while also fostering alignment and support among employees [61]. Both internal enablers (such as organizational structure, skills development, and cultural change) and external enablers (such as collaborative partnerships) are crucial to the success of the transformation. In other words, a digital transformation initiative cannot function in isolation, nor can a project be executed independently of the broader organization [61].

Thus, the process of digital transformation in HEIs involves a combination of technological and cultural changes. This not only alters the educational delivery models of universities but also redefines student expectations [80,82,83]. From an HEI management perspective, digital transformation strategies aim to (a) increase total revenue, (b) improve productivity, (c) generate value through innovative practices, and (d) build a reputation for brand and innovation [81,84]. Additionally, a sustainable digital transformation strategy must address the rapid challenges in the business environment, such as (a) virtualization and cloud computing, (b) integration of Wi-Fi technologies, (c) reduction of electronic waste, (d) energy efficiency, and (e) architectural infrastructure [67]. Incorporating these elements into the digital transformation plan ensures long-term viability [71].

Table 8
Challenges of digital transformation in the context of HEI management.

Group	Description	Authors
1. Structural elements	Key elements for a digital transformation vision include leadership, digital skills and technological knowledge, in addition to facing challenges such as resistance to change and complexity in systems integration.	[10,21,70]
2. Management and strategy	Managerial, strategic and educational difficulties in adopting AI, with emphasis on the high costs and complexity of digital transformation. Alerts for negative impacts on profitability, efficiency, privacy and data security.	[11,21,71]
3. Implementation	Includes studies that focus on challenges related to the implementation of digital transformation initiatives, seeking to improve services through standardization and automation.	[22,58,67]
4. Skills	Clarifies challenges related to the shortage of digital skills among employees, teachers and students, as well as the lack of adequate technical support.	[23]

Source: Prepared by the authors.

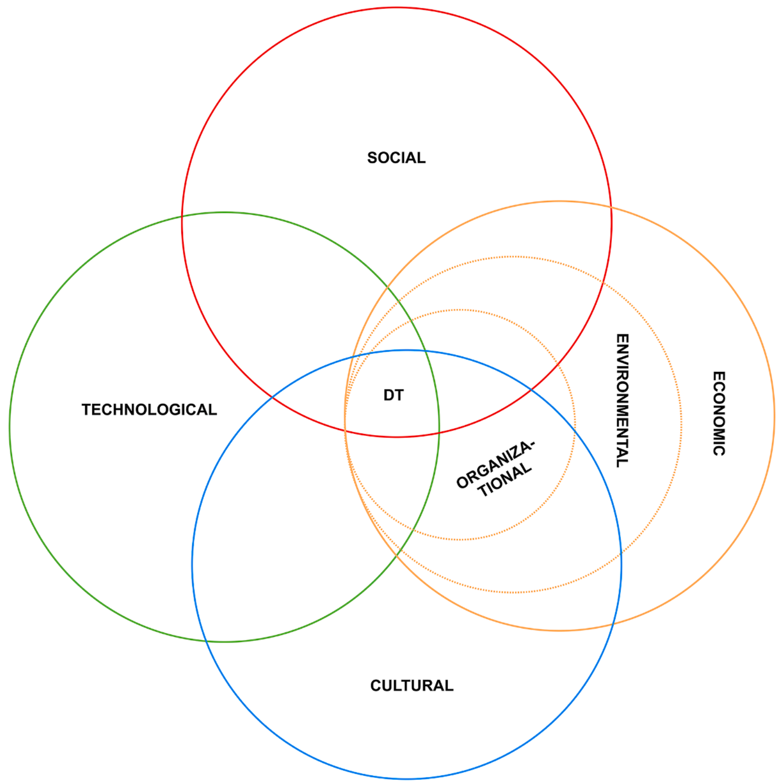

Fig. 5. Interpolation of the dimensions of analysis of digital transformation in the context of management of higher education institutions.

From an operational perspective, the success of digital transformation in HEIs is intrinsically linked to their ability to continuously generate and capture relevant information, such as student engagement in academic activities, achieved outcomes, and satisfaction with the services provided [10]. Based on the reviewed literature, it is clear that the critical success factors for digital transformation in HEI management include Culture and Leadership, Strategy and Planning, Management and Processes, Technology and Infrastructure, Human Capital, Partnerships and Collaboration, Sustainability, and User Experience. These factors must be integrated holistically into the strategic plans of HEIs, forming a significant foundation for addressing challenges and realizing the benefits of digital transformation.

Fig. 6 outlines the main critical success factors for digital transformation in HEIs. Its goal is to provide an overview of the practices identified in the reviewed literature, starting with the definition of vision and objectives and extending to continuous evaluation and feedback. Setting clear objectives is important for guiding efforts and ensuring that digital transformation aligns with the mission and values of the HEI [18,85].

After defining the vision and objectives, the HEI should conduct a thorough analysis of both internal and external contexts. The internal analysis should examine organizational culture, technological infrastructure, and available human resources. The external analysis should focus on trends in the educational market, student demands, and stakeholder needs, all of which should be continuously monitored and evaluated. HEI management should track the progress of the transformation through performance indicators and stakeholder feedback, adjusting strategies and action plans based on evaluation results to ensure the transformation's success in subsequent stages.

Nine studies have been identified that provide guidelines for digital transformation processes in HEIs. These studies are divided into two groups. The first group, comprising six articles, offers guidelines for transformation processes, addressing everything from the digitalization of educational activities to the sustainability of the transformation [11,67,77,86]. These studies emphasize the importance of process evaluation, strategic planning, and the analysis of pedagogical, technological, and organizational contexts. The second group, consisting of three articles, focuses on measuring transformation, defining parameters for assessing the digital readiness and maturity of HEIs through studies that range from readiness for change to evaluating organizational proficiency in digital transformation [57,69,87].

The literature reveals a lack of a structured approach to digital transformation in universities [72]. Given the diversity of operational models in different contexts and objectives, the results vary across the studies examined. However, the work dimensions identified in each model (Table 9) can serve as a foundation for developing strategies to implement digital transformation in HEIs. Notably, the dimensions associated with "Capacity Building," "Planning and Strategy," "Governance and Finance," and "Digital and Organizational Culture" consistently appear in most models, underscoring their importance in the transformation process.

Table 9 reveals that certain dimensions emerge as essential for digital transformation in HEIs, though they vary in specificity. This diversity suggests that the examined studies do not fully address all relevant dimensions. Additionally, the literature review highlights attempt to provide models for digital transformation strategies in HEIs; however, these models tend to be superficial and often lack substantial detail. This indicates that the existing literature does not offer a comprehensive approach to the stages of digital transformation—nor the specific methods and tools required for each stage within HEI management. This gap points to a significant research opportunity, as implementation remains the most tangible phase of transforming HEIs [88]. Thus, the need to explore and develop specific strategies for achieving digital transformation within HEI management presents both a challenge and a fertile area for further research.

Discussions around implementing digital transformation in HEIs are evolving in two main areas. On one hand, models are being developed for integrating technology into university infrastructures. On the other hand, recommendations are being proposed to measure the readiness and maturity of HEIs in adopting technology. digital transformation in

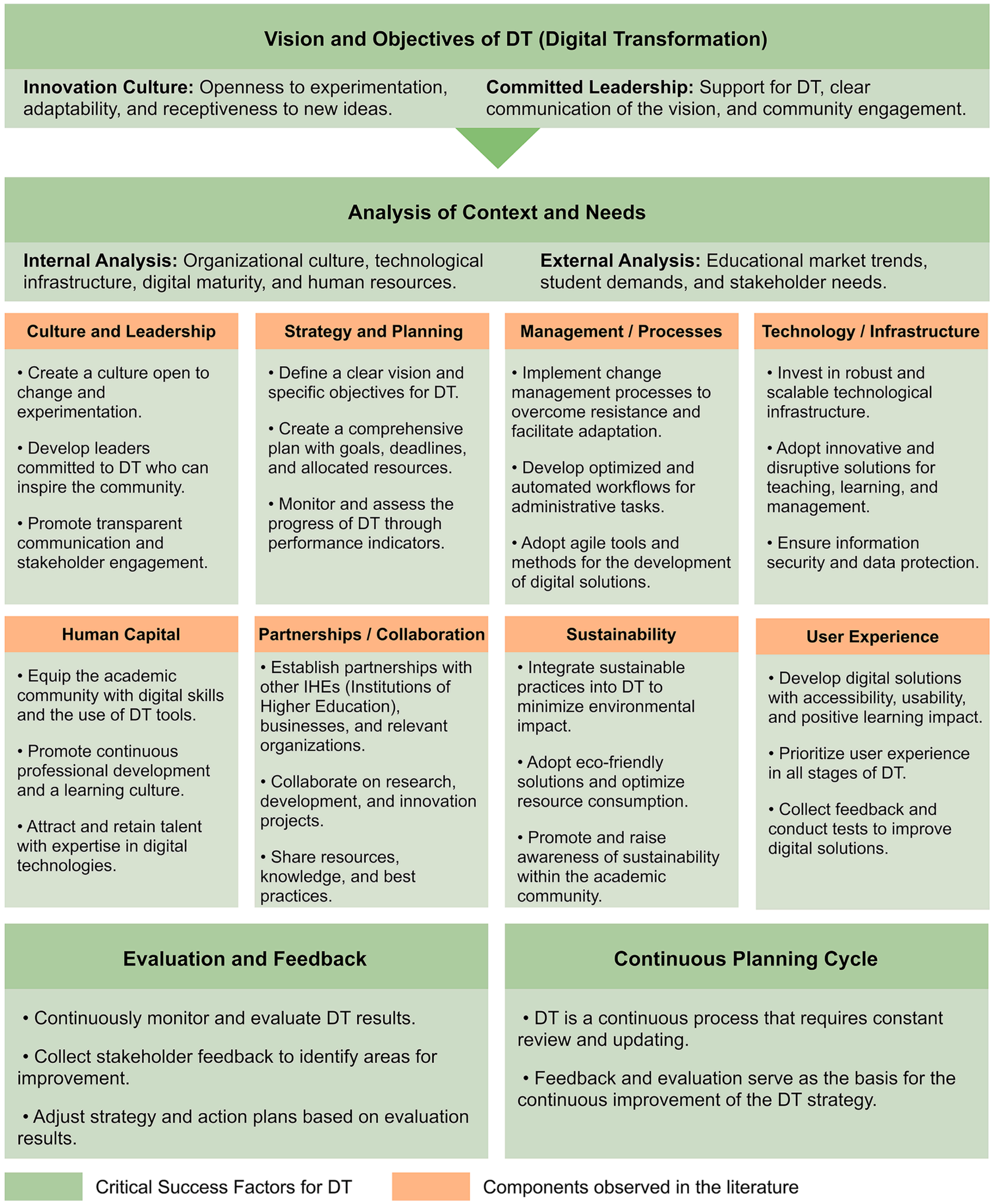

Fig. 6. Critical factors for the success of DT in HEIs.

HEIs is approached from various perspectives, and a consensus on its application has yet to be reached [17].

5. Conclusions

Through a systematic literature review, this study offers a comprehensive and nuanced understanding of digital transformation in HEI management, bringing together significant contributions from various perspectives in the literature. The detailed analysis of the corpus revealed the intricate interconnections among technological, organizational, social, economic, environmental, and cultural dimensions, deepening our understanding of the phenomenon. The study identified key enabling technologies, and the associated challenges and obstacles related to digital transformation in HEIs. Notably, Artificial Intelligence, Cloud Computing, and Big Data emerged as the most relevant enabling technologies, indicating an initial adoption trajectory. However, the diversity of approaches across the studies highlights a substantial gap in the literature, particularly regarding the implementation stages of digital transformation in HEI management.

The identification of barriers and challenges, categorized into "Structural Elements," "Management and Strategy," "Implementation," and "Skills," provides a detailed view of the complexities institutions face during this process. The varying perspectives on digital transformation in higher education, predominantly focused on academic and pedagogical approaches, reveal a gap, particularly in its role as a driver of transformation in institutional management and competitiveness.

Understanding digital transformation as a catalyst for change in institutional business models points to a significant reconfiguration of products, services, and operations, promoting efficiency compared to traditional practices. The ongoing transition from the third to the fourth

Table 9
Dimensions present in the digital transformation models of HEIs.

Article	Governance and finance	Planning and strategy	Training	Digital and Organizational Culture	Entrepreneurship and innovation	Teaching-learning	Leadership	Technology and infrastructure	Research	Evaluation	Communication	Change management	Social projection
[86]	•	•	•			•			•				
[11]										•	•		
[77]	•	•				•		•					
[61]	•	•	•	•	•							•	
[62]	•	•	•	•			•	•					
[73]	•				•				•				•
[59]	•		•			•							
[87]		•	•	•			•	•					
[69]		•		•	•		•						
OCCURRENCES	*6*	*6*	*5*	*4*	*3*	*3*	*3*	*3*	*2*	*1*	*1*	*1*	*1*

Source: Prepared by the authors.

model of higher education institutions, marked by the digitalization of educational and management activities, presents both intriguing challenges and opportunities for improvement. However, the observed lack of a detailed roadmap for implementing digital transformation in higher education management underscores the urgent need for future research. Additionally, elements such as leadership, human capital, experience, and technology emerge as fundamental, providing a strategic foundation for the successful integration of digital transformation into institutional management.

By identifying key enabling technologies and structural and strategic challenges, this study emphasizes the need for a detailed roadmap for implementing digital transformation within HEIs. Moreover, by examining the lack of uniform approaches in existing literature, the study serves as a starting point for future research, offering valuable insights for developing practical, evidence-based strategies aimed at enhancing the management and competitiveness of HEIs.

Finally, this study contributes to identifying critical success factors for digital transformation in HEI management, highlighting the inherent complexities and opportunities within this sector. Future research could explore the evolution and impact of enabling technologies over time in institutions. Longitudinal analyses could reveal how these technologies influence long-term outcomes in terms of efficiency, innovation, and competitiveness in HEIs.

Thus, the main contribution of this study lies in the identification and systematization of critical success factors for digital transformation in HEI management, focusing on three core aspects: enabling technologies, challenges and barriers, and gaps in the literature concerning the implementation of digital transformation in institutional settings. The analysis highlights the interconnections between technological, organizational, social, economic, environmental, and cultural dimensions, demonstrating how these factors shape the adoption and integration of digital transformation in HEIs management.

Regarding the limitations of the present study, four were identified. The first limitation lies in the fact that some relevant literature may have been omitted, as, in any Systematic Literature Review, there is no guarantee that all existing academic publications were included. The second limitation stems from the exclusive use of the databases specified in the protocol, which may have practically restricted the scope of the search. A third limitation stems from a potential bias associated with the use of bibliometric data, which may inadvertently favor widely cited articles at the expense of lesser-known works that are nonetheless capable of offering equally relevant contributions. Lastly, the rapid evolution of digital technologies introduces a temporal constraint, as the landscape can change significantly between the data collection period and the publication of the findings, affecting the applicability of the results.

In future research, the scope of the search could be expanded to include additional databases, ensuring that relevant academic works possibly overlooked in the current review are captured. Moreover, a study focused on examining the practical application of the identified models and theoretical frameworks would be beneficial, comparing the real-world implementation of digital technologies in diverse institutional contexts with the conceptual propositions found in the existing literature.

Our paper addresses the pivotal issue of how emerging digital technologies are reshaping the strategic and operational management of higher education institutions (HEIs). Technologies such as Artificial Intelligence, Cloud Computing, and Big Data have been shown to improve efficiency and foster innovation in HEIs. However, there remains a need for clarification regarding the specific barriers and critical success factors involved in this transformation process. Through the Literature Grounded Theory method, we conducted an extensive analysis of the existing literature, presenting a comprehensive framework that identifies key enabling technologies and the associated challenges. Our findings are significant as they provide a structured overview of the technological landscape, addressing gaps in the literature, particularly in the implementation phases of digital transformation in HEI management.

Given that our study focuses on the strategic management of HEIs through the lens of digital transformation, we believe it is an excellent fit for publication in The International Journal of Management Education. We are confident that our study's findings will be highly relevant to your readership, offering valuable insights for improving the competitiveness and management effectiveness of HEIs, which aligns closely with the journal's focus on management education and innovation.

The article "Digital Transformation in the Management of Higher Education Institutions" is an original work that has not been previously published, and it reflects independent research conducted by the authors. We also confirm that none of the authors have any conflicts of interest related to this study.

Thank you for your time and consideration.

Sincerely,

CRediT authorship contribution statement

Jose Ermeson Silva Carmo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daniel Pacheco Lacerda:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cristina Orsolin Klingenberg:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis. **Fabio Antonio Sartori Piran:** Data curation, Methodology, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this article, **"Digital Transformation in the Management of Higher Education Institutions."** The authors have no financial, personal, or professional relationships that could be perceived as having influenced the research, analysis, or conclusions of this study. All affiliations mentioned are purely academic, and the research was conducted independently without any external pressures or interests from third parties.

APPENDIX A

Table 10

Table 10
Research Protocol.

Research Protocol			
Research Title: DIGITAL TRANSFORMATION IN HIGHER EDUCATION MANAGEMENT			
Work Team: *Omitted for Blind Review*			
Stakeholders: Academy and experts			
Revision: 001	**Date:** 07–28–2023	**Reviewed by:**Experts	
1. Research Question:			
a) What theoretical assumptions support the implementation of digital transformation in higher education management?			
b) What are the antecedents of digital transformation in higher education management?			
c) What are the stages of digital transformation in higher education management?			
d) What are the organizational domains/contexts orchestrated by digital transformation in HEIs?			
e) What are the enabling technologies of digital transformation for higher education management?			
2. Research Objective(s):			
a. Map the theoretical assumptions that support digital transformation in the management of higher education institutions (HEIs), positioning the literature on the topic.			
b. Investigate the relationship between technologies and digital transformation in the context of HEIs.			
c. Identify to what extent there are studies that support the implementation of technologies for digital transformation in the management of HEIs.			
3. Scope of Review:			
3.1 Breadth:	() Restricted	(**X**) Wide	
3.2 Depth:	(**X**) Superficial	() Deep	
3.3 Type of Review:	() Aggregative	(**X**) Configurative	
4. Theoretical Framework:			
The adoption of digital technologies has led to profound changes in higher education institutions (HEIs) across all dimensions, making it a critical priority for maintaining competitiveness in the 21st century [17]. In this context, HEIs are navigating disruptive scenarios driven by shifts in business models that directly affect their relationships with internal and external stakeholders, society, and governments [89]. Despite the development of robust strategies for digital technology implementation, many universities still lack the strategic vision, capacity, and commitment necessary for effective execution [90]. This underscores the need for further investigation in the specialized literature on the stages of the digital transformation process in HEIs, particularly focusing on how digital technologies are integrated into management to enhance operations. Additionally, in a business context, this may involve the introduction of new products, services, or a reconfiguration of the business model. The examination of pedagogical aspects and their outcomes is beyond the scope of this research.			
5. Time Horizon:			
No time filter			
6. Search String:			
("digital transformation") AND ("education" OR "college" OR "university" OR "school")			
7. Search Source:			
Web of Science, SCOPUS and EBSCO *Host*.			
8. Search Approach:			
(**X**) Direct search	(**X**) Contact with experts	(**X**) Snowball	() Other:
9. Eligibility Criteria:			
9.1 Inclusion criteria:	1. Documents in English.		
	2. Full texts.		
	3. Peer-reviewed scientific articles.		
	4. Documents whose central theme is digital transformation.		
	5. Open Access documents.		
9.2 Exclusion criteria:	1. Inaccessibility of the document.		

(*continued on next page*)

Table 10 (*continued*)

Research Protocol			
	2. Duplicate documents.		
	3. Research that does not address digital transformation as a central theme.		
10. Data analysis:			
10.1 Scientometric Analysis	**(X)** Scientific development		
10.2 Bibliometric Analysis	() Search performance	**(X)** Scientific mapping	
10.3 Content Analysis	() Aggregative	**(X)** Thematic analysis	() Structural analysis
11. Data summary:			
11.1 Aggregative synthesis:		() Quantitative meta-analysis	**(X)** Qualitative meta-analysis
11.2 Configurative synthesis:		**(X)** Meta-synthesis	() Other:

Source: Ermel et al. (2021).

APPENDIX B

Table 11

Table 11
Selected primary studies.

Ref.	Title	Authors
D01	Categories for barriers to digital transformation in higher education: An analysis based on literature	[70]
D02	Methodological to digital transformation in the strategic development of a university	[91]
D03	Deep Dive into digital transformation in Higher Education Institutions	[10]
D04	The challenges of instructors' and students' attitudes in digital transformation: A case study of Saudi Universities	[21]
D05	Digital transformation challenges: strategies emerging from a multi-stakeholder approach	[71]
D06	Education and digital transformation: The "Riconnessioni'' Project	[11]
D07	Digital transformation initiatives in higher education institutions: A multivocal literature review	[22]
D08	Change of the higher education paradigm in the context of digital transformation: From resource management to access control	[58]
D09	Digitalization of Education: Models and Methods	[77]
D10	A sustainable University: digital transformation and Beyond	[3]
D11	Higher Education Strategy in digital transformation	[67]
D12	Mastering the digital transformation Process: Business Practices and Lessons Learned	[61]
D13	digital transformation of University Education in Ukraine: Trajectories of development in the conditions of new technological and economic order	[92]
D14	Stages of digital transformation of Educational Institutions in the System of Sustainable Development of the Region	[93]
D15	Digital transformation in higher education: A case study on strategic plans	[86]
D16	Factors Influencing the digital transformation of Universities in Thailand	[62]
D17	Building an Integrated digital transformation System Framework: A Design Science Research, the Case of FedUni	[79]
D18	digital transformation Challenges for Universities: Ensuring Information Consistency Across Digital Services	[6]
D19	Digital transformation in education: Critical components for leaders of system change	[94]
D20	Current trends in the digital transformation of higher education institutions in Russia	[23]
D21	Higher Education Leadership in a Time of Digital Technologies: A South African Case Study	[85]
D22	Exploring the Readiness for digital transformation in a Higher Education Institution towards Industrial Revolution 4.0	[59]
D23	Same, but Different? digital transformation in Swiss Vocational Schools from the Perspectives of School Management and Teachers	[87]
D24	Assessing digital transformation in Universities	[69]
D25	digital transformation for Business Model Innovation in Higher Education: Overcoming the Tensions	[63]
D26	Digital transformation of a university as a factor of ensuring its competitiveness	[95]
D27	Production of indicators for evaluation of digital transformation of modern university	[76]
D28	University e-Readiness for the digital transformation: The Case of Universidad Nacional Del Sur	[20]
D29	A Systemic Perspective for Understanding digital transformation in Higher Education: Overview and Subregional Context in Latin America as Evidence	[78]
D30	How Higher Education Institutions Are Driving to digital transformation: A Case Study	[96]
D31	An Institutional Perspective for Evaluating digital transformation in Higher Education: Insights from the Chilean Case	[72]
D32	digital transformation at the University of Porto	[97]
D33	University digital transformation Intelligent Architecture: A Dual Model, Methods and Applications	[73]
D34	Promising digital university: a pivotal need for higher education transformation	[15]
D35	digital transformation: Insight from Leaders in the Mid-rank Universities in Indonesia	[18]
D36	Intelligent Educational Dual Architecture for University digital transformation	[98]

Source: Prepared by the authors.

Data availability

No data was used for the research described in the article.

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